

SAGAR SAHU

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EDUCATION

Shri Shankaracharya Institute of Professional Management & Technology

Bachelor of Technology in Computer Science & Engineering; CGPA: 7.3/10.0

Raipur, CG

2023 – 2027

EXPERIENCE

AI/ML Vocational Trainee

Jun 2025 – Aug 2025

IIIT Naya Raipur

Naya Raipur, India

- Engineered a hypertension screening system [Source Code] achieving **87.88% accuracy** on **25,201 PPG signals** by implementing **PPG waveform feature extraction** and **XGBoost** classifiers.
- Attained **93.84% specificity** for clinical viability by utilizing **GroupShuffleSplit** for subject-independent validation, successfully preventing data leakage.
- Boosted model performance by **31%** (67% to 87.88%) over baseline by optimizing **frequency-domain heart-rate variability features** (LF/HF ratio) and addressing class imbalance with **SMOTE**.

AI-ML Virtual Intern

Oct 2025 – Dec 2025

Google for Developers (MoE & AICTE Partnership)

Remote

- Achieved **Grade O (Outstanding: 90-100%)** in a competitive 10-week AI-ML internship sponsored by Ministry of Education and supported by Google for Developers through the India Edu Program.
- Built a strong working foundation in deep learning using **TensorFlow** across 6 modules: neural network programming, object detection with **bounding box regression**, and **CNN architectures** for image classification.
- Learned the fundamentals of **visual search engine** development, including similarity scoring approaches and connecting image databases to ML models for product image retrieval.

PROJECTS

Twitter Sentiment Analysis System | Python, DistilBERT, HuggingFace, Streamlit | Website | Source Code

Jul 2026

- Built an end-to-end sentiment analysis system by fine-tuning **DistilBERT** on **45,000 tweets**, achieving **~72% validation accuracy** – within the published benchmark range for the dataset.
- Implemented a custom text preprocessing pipeline handling emoji normalization, slang expansion, stopword removal, and multilingual auto-translation.
- Deployed as an interactive **Streamlit** web app supporting single-text and batch CSV analysis with real-time confidence visualization.

Mechanic Mitra: Enterprise Multi-Agent System | Gemini 2.5, Agentic Workflows | Source Code

Nov 2025

- Built a **multi-agent system** with 4 coordinated agents (Visual, Audio, Pricing, Chief) to streamline vehicle diagnostics from a typical 45-minute manual inspection to a 3-minute automated assessment.
- Implemented a persistent **DiagnosisMemory** component that improves classification accuracy over time by tracking patterns in historical fault data.
- Added resilience with **exponential backoff** and error handling to manage API rate limits (15 RPM) for the **Gemini 2.5** model.

PDF Summarizer | Python, Gemini Flash Lite, Streamlit | Website | Source Code

Oct 2025

- Reduced document analysis time by **85%** for end-users by engineering a summarization engine using **Gemini 2.5 Flash Lite**.
- Architected an intelligent caching layer to maximize free-tier API quotas, enabling sustained operation within **1,500 requests per day** capacity at zero cost.
- Ensured **99.9% uptime** for file uploads up to 5MB by managing session states securely on **Streamlit Cloud**.

SKILLS

Generative AI: Vertex AI, Gemini 2.5, Prompt Engineering, Multi-modal Agents

Machine Learning: XGBoost, Scikit-learn, Pandas, NumPy, HuggingFace Transformers, Signal Processing, Bio-medical Analysis

Cloud Infrastructure: Oracle OCI (Certified Professional), Google Cloud Platform (15 Verified Skill Badges)

Development: Python, SQL, Git, Streamlit, REST APIs, JSON Data Handling

CERTIFICATIONS

Oracle University: OCI 2025 Certified Generative AI Professional (**Oct 2025**) | Data Science Professional (**Oct 2025**)

Google Cloud: Generative AI Fundamentals (Udacity) | 15 Google Cloud Skill Badges (Infrastructure & AI)

Simplilearn: Machine Learning using Python (**Aug 2025**) | Python for Beginners (**Sept 2025**)

Industry: Data Analysis Using Pandas (Vodafone/_VOIS) | IP Strategy (L2Pro India/Qualcomm)